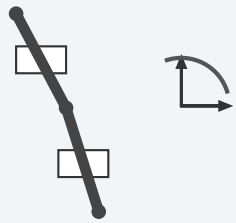
Sparse IMU gait sensing



Sensor-frame rotation
changes axis channels
0-15 deg stress test

Virtual benchmark 10-deg rotation

Unaugmented
8.08-8.24 deg RMSE

10-deg augmentation
3.66-4.27 deg RMSE

Frame registration
3.19-3.92 deg RMSE

Augmentation passes the 10-deg gate
No cap passes 12.5/15 deg

Physical-IMU test 10-deg rotation

Frame rotation
degrades 4/4 endpoints

Augmentation
improves 3/4 endpoints

Registration
improves 4/4 endpoints

Physical-IMU signals;
frame rotations imposed
mathematically